Stanley Nelavala

Software Engineer: Backend Services, Optimization and Fast Data Paths

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Huntington, IN | Open to relocate anywhere

Summary

Software engineer with an M.S. in Computer Science (4.0 GPA) and a year at Mass Technologies, a 10-person startup, where I build the RxApps360 hospital platform: FastAPI and PostgreSQL services, React and TypeScript web apps, a React Native mobile app, and the signed API every module calls. I care about correctness and speed: keyset paging took a query on a 328,000-row catalog from 285 ms to 0.20 ms. Outside work I build tested public services in Python, Go, Rust and Java, and I use Claude Code, Codex and opencode every day. AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test) *Huntington, IN*

- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Replaced offset paging with keyset paging on the catalog device list: row 150,000 of a 328,000-device dev database fell from 285 ms to 0.20 ms, with no skipped or repeated rows (tested with inserts, deletes and ties).
- Built RxDesign, the hospital design service (FastAPI, React, TypeScript), with a deterministic, rule-based layout engine and an LLM setting; 220+ offline API tests run with fake models, so tests never call an LLM.
- Built the RxAsset iOS and Android app (React Native, Expo) with scan-to-search and an on-device voice assistant (speech runs on the phone, no cloud call); set up release pipelines in GitHub Actions with automatic TestFlight submission.
- Automated weekday start and stop of dev and acceptance servers and databases (ECS, RDS): about 53 hours a week instead of 168.
- Standardized malware scanning of file uploads (GuardDuty) across 9 app modules.
- Use Claude Code, Codex and opencode every day to design, write, debug and review code; tests and written acceptance checks gate every change.

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR, BERTopic) that analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation.

AI projects (public code, tests)

Inference Gateway | *Go, gRPC, ONNX Runtime, PyTorch, OpenTelemetry, Helm* GitHub

- Go service for an int8-quantized classifier with micro-batching, deadlines and load shedding; macro-F1 0.8636 for fp32 and int8; sized for a 10x peak at 2 to 3 replicas (local benchmark).

Agent Integration Platform | *Python, FastAPI, LangGraph, MCP, PostgreSQL (pgvector), Redis* GitHub

- Multi-step agent with typed state, one constrained MCP write tool, human approval that resumes after restart, and step and token budgets; hybrid retrieval with access labels.
- 62 deterministic replay evals, prompt-injection tests, output guardrails, PII redaction, an append-only audit log; failover between two LLM adapters with a circuit breaker.

Payments Ledger Service | *Java, Spring Boot, PostgreSQL, React, TypeScript* GitHub

- Double-entry ledger with idempotency keys and an outbox; property tests and a load check that no money is created or lost.

Skills

Role focus: low-latency data paths, query and paging optimization, backend services at scale, tested code
AI-assisted coding: Claude Code and subagents, Codex, opencode, a DeepSeek harness; planner, manager and worker runs with acceptance checks
Languages: Python, TypeScript, JavaScript, Java, Go, Rust, SQL
Backend and web: FastAPI, React, React Native (Expo), Node.js, PostgreSQL, Redis, REST, gRPC, keyset paging
Cloud and quality: AWS (ECS, RDS, S3, GuardDuty), Docker, GitHub Actions, pytest, Playwright, OpenTelemetry

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)